Observer Patch Holography
A Compact Case for an Observer-First Unification Program

Paper release: r2001 Released: July 27, 2026

Abstract

Observer Patch Holography starts with finite self-reading systems that expose boundaries, keep records, compare overlaps, and repair disagreement. The same architecture supports exact public-record and event-algebra results, Lorentz kinematics on a spherical screen, a conditional Einstein composition, and an icosahedral coefficient algebra with the Standard Model Lie type. Deterministic simulations measure a Lorentzian event form with one time and three space directions across a support-adjusted three-rung path. Interval arithmetic certifies unique roots of the declared pixel maps. A positive-chamber circulant theorem gives the Koide relation exactly, while finite capacity identities supply a conditional de Sitter time-advance mechanism. The physical attachments are explicit. A sorry-free Lean library, exact code receipts, and reproducible simulations make the case testable at each step.

Observer Patch Holography (OPH) asks whether physics can be reconstructed from the minimum machinery required for a public world. A patch has local state, a boundary, records, readback, and repair moves. Each patch sees a fragment. Agreement across shared boundaries turns private descriptions into public facts. The selected world is a stable normal form,

$$T(\mathfrak{U}) = \mathfrak{U}.$$

The theory rests on three core axioms.

The three axioms

1. Oriented twelve-port observer screen. There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron; carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Formally: for every regulator r there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$ whose carriers carry twelve primitive central port projections and the exact boundary packet $K = (P, E, F, o)$, joined by seam algebras with coherent triple-overlap cocycles into a nerve N_r with a degree-one bridge b_r to the oriented spherical support S_r, all commuting with refinement.

2. Observer agreement. Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally: the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map on accepted public data.

3. Conditional maximum randomness. Everything that observer agreement leaves unconstrained is maximally random. Formally: the realized state is the information projection of an exact reference family onto the convex set of compatible local state families satisfying the finite observer-visible constraints. The finite A1-generated observer cover is state-determining on that feasible set, and its exact weights are strictly positive: $\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_P w_{r,P} D(\rho_{r,P} \| \tau_{r,P})$.

The first axiom constrains architecture, the second meaning, the third state selection. None of them contains a gauge group, a particle list, a recovery law, or a rule that fixes field content or multiplicity. Every result below states which of these axioms it uses and which further premises it carries: an exact theorem over all models, an exact result inside one named finite realization, a discovery-level observation, a declared open interface, an independence result with countermodels, a physical identification, or a withdrawn claim.

The proposal earns attention because several familiar structures arise from this small basis. They also arrive with different evidence types. Exact theorems, interval certificates, conditional physical compositions, and simulation measurements occupy separate evidence classes throughout this paper.

The claim in one sentence

Finite observer-like patches provide a common construction for public quantum records, Lorentz kinematics, a conditional Einstein branch, compact gauge structure, matter representations, and quantitative closure tests. The mathematics fixes a substantial skeleton. Physical carrier identifications determine how much of that skeleton describes our universe.

1. Nine receipts that carry the case

The following count contains the strongest independent pieces of evidence. Every row states its boundary in the same breath as its result.

	Receipt	Closed content	Evidence
1	Public records and finite event algebra	Terminating repair gives protected normal forms on the declared confluent carrier class. The completed central record surface obeys Born probabilities, Lüders conditioning, and the Tsirelson bound.	Lean and exact finite code [1,2,10]
2	Lorentz kinematics and three observer-frame dimensions	On the certified spherical support branch, $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ and $\dim[\mathrm{SO}^+(3,1)/\mathrm{SO}(3)] = 3$. Event population is a separate physical receipt.	Theorem chain [1,3]
3	Einstein composition and measured cone emergence	The typed modular, null-stress, entropy, and small-ball premises compose to the Einstein relation. A support-adjusted three-rung path measures inertia $(1,3)$ with cone margins $-5.62, -3.22, -1.41$. A same-size support-width control changes the inertia to $(2,2)$.	Lean implication, instruments, simulation [3,11]

	Receipt	Closed content	Evidence
4	Icosahedral Standard Model Lie type	The twelve-port coefficient module carries an exact compact bracket of type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. A physical current lift is an explicit additional premise.	Lean, exact algebra, code [3,10]
5	Global quotient and one-generation representation witness	Trace balance gives $S(U(3) \times U(2))$, the shared center gives $\mathbb{Z}_6$, and an exterior algebra branches into one fifteen-state Standard Model generation with the anomaly arithmetic. Among the single complete faithful screen bands in $3 \leq N_g \leq 5$, the exact Laplacian cost selects rank three. Physical matter-pole and family identification are open.	Lean and exact code [3,10]
6	Pixel fixed-point arithmetic	Outward-rounded interval certificates prove existence and uniqueness for each declared pixel map and exclude a second root across its analytic domain. The fine-structure comparison is diagnostic because the physical transport map is incomplete.	Interval arithmetic [5,10]
7	Charged-lepton interval diagnostic	The measured electron, muon, and tau masses lie inside outward-rounded transport enclosures. Their logarithmic half-width is 1.732%, giving one-sided multiplicative widths of −1.72% and +1.75%. Target-anchored ratios and empirical transport make this a closure diagnostic.	Interval code and ledger [6,10]
8	Positive-chamber Koide theorem	A Hermitian three-cycle response obeys $Q = \frac{1}{3} + \frac{2}{3}(\lvert b\rvert/a)^2$. Hence $Q = 2/3$ exactly at $\lvert b\rvert/a = 1/\sqrt{2}$. Equal finite event blocks supply this balance under the declared tracial packet. Under the balance and ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure, 0.43σ from measurement. Phase and physical family attachment are open.	Lean and exact code [7,10]
9	Finite de Sitter identities	Pure de Sitter normalization, the finite entropy maximum, the exact capacity transfer law, its analytic curvature, and the port/edge line-graph identity are exact. The gravitational shock sign requires the stated horizon and observer dictionaries.	Lean and exact code [8,10]

The repository contains more than 900 public Lean theorem and lemma declarations. The count includes positive results, premise boundaries, and countermodels. It is useful because the formal statements expose assumptions that prose can otherwise hide. The finite receipts bind inputs, outputs, tolerances, and negative controls to reproducible artifacts [9].

2. Public reality and quantum records

Take a finite family of patches x_i, each with an observable boundary map. For every overlap $e = (i, j)$, the two induced records must agree. A repair move changes local state while preserving protected readout. On a terminating, frustration-free carrier with a confluent accepted repair relation, every schedule reaches the same quotient normal form. Exact code exhausts all six declared overlap schedules on the reference carrier and returns the same normal form [2,10]. The general confluence hypothesis is part of the carrier contract. Arbitrary local repair does not imply it.

Once a compare, write, and verify slice has completed, its accessible events generate a finite central record algebra. If P_E is the projector for an event E and ρ is the normalized state, then

$$\Pr(E) = \mathrm{Tr}(\rho P_E), \qquad \rho \mid E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

The first expression is the Born probability on this finite surface. The second is Lüders conditioning. The corresponding Clauser–Horne–Shimony–Holt (CHSH) parameter satisfies

$$\lvert S_{\mathrm{CHSH}}\rvert \leq 2\sqrt{2}.$$

These identities belong to a 112-declaration audited finite event-algebra development [1,10].

The result gives quantum record arithmetic from public observability. It does not construct an arbitrary continuum quantum field theory. Its value is more specific: a finite observer contract produces the probability and update rules required by physical branches built on it.

3. Lorentz kinematics and the Einstein branch

An oriented conformal spherical support has the connected symmetry

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1).$$

The space of unit timelike observer frames is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 6 - 3 = 3.$$

This is an exact kinematic result on the spherical support branch. Local icosahedral carrier geometry, the federation nerve, and the global S^2 support are distinct objects. A populated four-dimensional event base requires record separation, local charts, an open-image condition, affine transition data, a quadratic cone, and causal reachability [3].

On one common algebra-state tower, normalized modular flow supplies a local clock, half-sided modular inclusions supply positive null translations, null tomography assembles local stress, and generalized-entropy stationarity with small-ball geometry gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle.$$

The Lean development proves the typed algebraic composition. A source-derived physical tower, the continuum limit, and the physical identification maps are premises [3,10].

The simulation branch tests the geometry produced by repair dynamics rather than inserting a Lorentzian metric into the readout. The selected path uses $(16{,}384, 128, 96)$, $(65{,}536, 256, 96)$, and $(262{,}144, 512, 384)$ for carrier count, observer count, and support width. The held-out event form has inertia $(1, 3)$ at every selected rung. Its cone margin moves through

$$-5.6, \qquad -3.2, \qquad -1.4.$$

Coupling spread decreases beside it. At 262,144 carriers, the support-width 96 control has 312 cross-observer edges and inertia $(2, 2)$; the support-width 384 row has 1,062 edges and inertia $(1, 3)$. Configurations, primary arrays, hashes, and regeneration scripts are public [10]. The data establish a reproducible support and cross-read sensitivity on the instrumented branch. They do not establish an invariant-density convergence law, an infinite-scale limit, or the missing cap-state thermal receipt.

Why this receipt is unusually strong
The observed signature is held out from the update rule, persists across three named configurations, changes under the same-size support-width control, and sits inside a typed theorem chain that states every physical premise. This combines an exact implication with a direct computational measurement.

4. From twelve ports to gauge and matter structure

The declared carrier lineage realizes the twelve ports with equal algebra traces, a realization property rather than axiom content, with oriented icosahedral incidence. Its proper rotation group is the alternating group A_5. Exact finite calculations give the distance profile, the antipodal pairing, the sixty proper rotations, the rank-three Gram frame, and the decomposition of the real port-coefficient space

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}.$$

Let six antipodal axes have unit vectors u_i, set $P_i = u_i u_i^{\intercal}$, and write $\Phi(b) = \sum_i b_i P_i$. The face orientation fixes the companion three-dimensional map. Pulling back the block commutator through the resulting equivariant isomorphism gives

$$(P_{12}, [\ ,\ ]_{\Theta}) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2).$$

This is an exact coefficient-space Lie algebra. The five-dimensional band is noncentral, the derived algebra has dimension eleven, and the center is the uniform even line. The source packet establishes the local incidence and coefficient construction on the declared carrier lineage. Physical promotion requires a full-rank compact current lift, refinement intertwining, and common source binding [3,10].

Trace balancing extends the same block construction to $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$. Its connected group is

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

The six-element kernel is exact, as is the independent lattice quotient with Smith invariants $(1, 1, 1, 1, 1, 6)$. Physical spin and deck descent are additional conditions.

The trace-balanced block $V = \mathbb{C}^3 \oplus \mathbb{C}^2$, with hypercharges $(-1/3, 1/2)$, has the exterior branching

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L.$$

It contains fifteen states, the Standard Model hypercharges for one generation, the expected three one-Higgs invariant lines, cancelled gauge and mixed anomalies, and exactly four weak doublets. This is a precise representation witness. Selecting that block as physical matter, excluding extra sectors, imposing spacetime spin statistics, and attaching three families require their own source receipts [3].

The screen coefficient module also gives an exact family-band selection under its named premises. Its four bands have ranks $1, 3, 3, 5$. Restricting to single complete faithful objects in the conditional window $3 \le N_g \le 5$, the Laplacian costs $5-\sqrt{5}, 6, 5+\sqrt{5}$ select the first rank-three band uniquely. The declared unitary simulator recovers that projector at its lowest positive generator frequency. Identifying this finite response mode with physical matter families requires the matter-pole, chirality, locality, and laboratory receipts.

The two exact routes meet at the same Lie type: transportable sectors support compact reconstruction, while the local twelve-port coefficient bracket constructs the algebra directly. Their agreement is useful evidence. A theorem of physical identity between the two current objects is a separate obligation.

5. Arithmetic closure and charged-lepton diagnostics

The local pixel fixed point

The local screen cell is asked to reproduce its readable electromagnetic width:

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

Here $A_T(P)$ is the inverse electromagnetic coupling emitted by a trial cell through the declared Thomson-limit map. For each declared map, outward-rounded interval Banach certificates prove existence and local uniqueness. A 256-piece derivative enclosure proves global at-most-one across the analytic domain, with an empty exceptional set [5,10]. The gauge-width map has the certified root

$$\alpha^{-1} = 137.035660136946577\ldots$$

beside the 2022 Committee on Data of the International Science Council (CODATA) value 137.035999177(21) [11]. The relative separation is 2.5×10^{-6}.

The arithmetic theorem is stronger than the physical reading. The comparison coordinate used by this map is anchored to the measured endpoint, and the source-derived same-scheme hadronic transport is absent. The result is a certified unique root of a declared map and a quantitative diagnostic. It is not an independent determination of the physical fine-structure constant.

The charged-lepton interval surface

The charged-lepton calculation combines a finite eight-path carrier with an empirical electromagnetic transport packet. Independent 100-digit decimal recomputation gives outward-rounded enclosures with logarithmic half-width 1.732%, corresponding to one-sided multiplicative widths of -1.72% and $+1.75\%$. The measured triple lies inside all three enclosures. A narrower conditional eight-register coordinate lies about 84 parts per million from the comparison triple [6,10].

Measured mass ratios and transport anchors enter this lane, so these numbers are diagnostics of internal closure. They give a sharp test surface without claiming a source-only mass prediction. Prospective endpoint status requires a physical charged determinant line, an absolute clock, and a source-emitted transport bridge.

An exact positive-chamber Koide theorem

Let $R^3 = I$ and consider the Hermitian three-cycle response

$$C = aI + bR + \bar{b}R^2.$$

When its three eigenvalues are nonnegative and are read as square-root masses, the normalized mass coordinate satisfies

$$Q = \frac{\sum_i m_i}{\left(\sum_i \sqrt{m_i}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2.$$

Consequently,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \frac{|b|}{a} = \frac{1}{\sqrt{2}}.$$

The phase cancels from Q and jointly controls the two mass ratios. Equal rank-two event blocks in the finite tracial Gelfand–Naimark–Segal packet give the required modulus balance exactly [7,10]. Lean checks the circulant identity and equivalence. Physical chiral-family attachment and phase selection are open, so the theorem is independent of the charged-lepton diagnostic above.

Under the balance and mass-ordering premises, the measured electron and muon masses determine the tau mass through one quadratic. The outward-rounded enclosure is [1776.968991, 1776.969063] MeV, a 72-eV window sitting 0.43σ from the measured 1776.93 ± 0.09 MeV, with the spurious root excluded below the muon mass and the premise ancestry declared: the balance condition was first abstracted from the measured triple, so this row is a conditional postdiction rather than a blind prediction. The window is three orders of magnitude narrower than the measurement uncertainty, and tau-mass programs in progress will move that uncertainty toward it: a future average outside the window refutes the balanced-circulant premise, while an average inside it passes a test the premise could have failed.

6. Fixed capacity and the de Sitter shock sign

For pure de Sitter space of radius L in spacetime dimension d,

$$r_c = L, \qquad \kappa = L^{-1}, \qquad \mu^2 = (d-2)\kappa r_c = d-2 = \lambda_{\ell=1}(S^{d-2}).$$

The radius and cosmological constant cancel. The smooth $\ell = 1$ modes are generated by de Sitter isometries and form the gauge kernel of the imported shock operator [8,13].

For finite sector dimensions d_i, total dimension $M = \sum_i d_i$, and sector probabilities p_i,

$$S_{\text{gen}}(p,d) = -\sum_i p_i \log p_i + \sum_i p_i \log d_i = \log M - D_{\text{rel}}(p\|d/M),$$

where D_{rel} is relative entropy. Thus $p_i = d_i/M$ gives the exact maximum $S_{\text{gen}}^{\max} = \log M$. If an observer receives a fraction f of a fixed horizon-observer capacity and the horizon sectors deplete uniformly, every admissible integer transfer obeys

$$\Delta S_{\text{gen}}^{\max} = \log(1-f) < 0.$$

The positive-real interpolation is strictly decreasing and concave. Its zero-transfer point is a one-sided boundary maximum.

The associated logarithmic sector observable has symmetric-point Hessian

$$\operatorname{Hess} \mathcal{A} = \frac{1}{nd^2}\left(I - \frac{2}{n}J\right).$$

Its fixed-total tangent curvature is positive, its homogeneous curvature is negative, and its symmetric-point gradient is nonzero. The one-sided capacity transfer carries the sign; an interior fixed-capacity extremum is excluded [8,10].

On the twelve-port icosahedral graph, the raw Laplacian spectrum and regular line-graph identity are exact. If the rotation triplet is an exact gauge sector and the physical shock kinetic term is the scaled nearest-neighbour port or edge Laplacian, normalization at that triplet gives

$$\{-2,\ 0,\ 1+3/\sqrt{5},\ 1+\sqrt{5}\}.$$

The edge carrier adds only eigenvalue 10 with multiplicity 18. Identifying capacity with horizon area, the transfer coordinate with observer mass, and the finite operator with the gravitational shock supplies a conditional time-advance mechanism. The finite identities make none of those physical identifications. They supply no positive cyclic static-patch trace. The obstruction of Chen, Stanford, Tang, and Yang excludes that trace proposal rather than the finite identities above [12].

7. Why the cumulative case is worth testing

Each receipt has a separate mathematical domain. OPH becomes interesting because the receipts share one finite architecture. Public records lead to the event algebra. Spherical support leads to Lorentz kinematics. Modular and entropy data lead to the Einstein composition. Icosahedral carrier data lead to the gauge coefficient algebra and an exact matter witness. The same finite readback idea produces the pixel closure, the tracial Koide balance, and the capacity transfer law.

The dependencies are visible enough to be attacked. A proposed carrier must emit local state, ports or boundaries, readback, records, repair moves, and a public evidence bundle. Physical claims require source-bound maps between the finite objects and their spacetime, current, matter, clock, or horizon interpretations. Lean checks symbolic implications. Exact programs check finite searches and interval arithmetic. Simulations measure branches whose continuum construction lies beyond the finite theorem.

Conditional continuations that fail their declared comparison gates are excluded from the evidentiary case in this paper. The audit ledger classifies those closed hypotheses separately from the exact record, geometry, gauge, Koide, and capacity results. Their failure does not strengthen the positive case, and none of their claims is used above.

The program therefore has a clear scientific shape. It contains exact mathematics with recognizable physical structure, a measured spacetime signal with mechanism controls, certified arithmetic closures, and explicit physical attachments that can succeed or fail. That combination supports a serious observer-first route toward unification. It does not amount to a completed physical theory of everything.

The shortest fair assessment

OPH has a substantial machine-checked core, several exact finite constructions that land on the algebraic skeleton of known physics, and a reproducible Lorentzian convergence signal. Its strongest physical conclusions are conditional on named carrier maps. This is a promising research program with unusually sharp boundaries and enough positive structure to justify direct work on the open attachments.

References

[1] Mueller, B., et al. (2026). "Observers Are All You Need."

[2] Müller, B., et al. (2026). "Reality as a Consensus Protocol."

[3] Müller, B., et al. (2026). "Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency."
"Deriving Standard Model Gauge Structure from Observer Overlap Consistency."

[4] Müller, B., et al. (2026). "Federated Echosahedral Screen Microphysics."

[5] Müller, B. (2026). "The Fine-Structure Constant as an OPH Pixel Fixed Point."

[6] Müller, B., et al. (2026). "Deriving the Particle Zoo from Observer Consistency."

[7] Müller, B. (2026). "The Positive-Chamber Koide Identity for Icosahedral Face Circulants."

[8] Müller, B. (2026). "The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity."

[9] FloatingPragma (2026). Lean library and proof index; exact code receipts and interval certificates.

[10] Mueller, B. (2026). Einstein-cone convergence evidence; OPH physics simulator and reproduction source.

[11] CODATA Task Group (2024). "2022 CODATA Recommended Values of the Fundamental Physical Constants."

[12] Chen, Y., D. Stanford, H. Tang, and Z. Yang (2026). "Negative Shocks versus Static Patch Holography," arXiv:2607.14042.